exctracting elemental iodine from an amazon product [WIP]

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lugol

hi! I've been wanting to make a relatively harmless and interesting extraction in my homelab, and I've realized Lugol (a solution containing iodine and potassium iodide dissolved in water) might be a good starting material. my objective is to separate the elemental iodine from the other components on Lugol, and ultimately obtain relatively pure elemental iodine (I_2) crystals at the end of my experiment.

starting material

Lugol's solution is an aqueous mixture that contians elemental iodine (I_2) and potassium iodide (KI). the iodine isn't present exclusively as isolated I_2 molecules in the aqueous solution. rather, iodide ions (I^-) supplied by the potassium iodide (KI) interact with molecular iodine (I_2) to form I_3^-, called triiodide: ***I2+I−⇌I3−***

why does it happen?

when **KI** (**potassium iodide**) dissolves in water, it separates into potassium ions (**K^+**) and iodide ions (**I^-**): **KI → K^+ + I^-**

the iodide ion can interact with an iodine molecule afterwards. I_2 has an electron-rich distribution, and I^- can interact with it to form the triiodide ion, I_3^-. the important consequence is that **I_3^- is much more comfortable existing in water than I_2 is**. because I_3^- is a charged species, water can stabilize it much more effectively than it can neutral I_2, because water is a polar molecule, as you may know. the iodide supplied by KI gives the iodine another, much more water-soluble form to exist as. this allows considerably more iodine to remain dissolved in the aqueous solution.